理论作业

Exercise 12.11

$u_t = vu_{xx} \ \in \omega := (0,1)\times(0,T)$

$$\begin{aligned}\frac{U_i^{n+1}-U_i^n}{k} &= \frac{1}{2}(f(U_i^n,t_n)+f(U_i^{n+1},t_{n+1}))\\ &= \frac{v}{2h^2}(U_{i-1}^n-2U_i^n+U_{i+1}^n+U_{i-1}^{n+1}-2U_i^{n+1}+U_{i+1}^{n+1})\end{aligned} \tag{1}$$

$$\begin{gathered}\Rightarrow U_i^{n+1}-\frac{kv}{2h^2}(U_{i-1}^{n+1}-2U_i^{n+1}+U_{i+1}^{n+1}) = U_i^n+\frac{kv}{2h^2}(U_{i-1}^n-2U_i^n+U_{i+1}^n)\\ (1-\frac{k}{2}A)U^{n+1} = (1+\frac{k}{2}A)U^n+b^n\end{gathered} \tag{2}$$

其中 $U^n=(U_1^n,U_2^n,\cdots,U_m^n)^T, b^n=\frac{r}{2}(g_0(t_n)+g_0(t_{n+1}),0,\cdots,0,g_1(t_n)+g_1(t_{n+1}))^T$.

Exercise 12.26

将空间离散化，得到 ODE 系统 $\frac{du}{dt}=Au, A=\frac{v}{h^2}D$,$\theta$-离散公式 $\frac{U^{n+1}-U^n}{k}=\theta\lambda U^{n+1}+(1-\theta)\lambda U^n$, 令 $z=k\lambda$, 有 $U^{n+1}=\frac{1+(1-\theta)z}{1-\theta z}U^n$.

令 $R(z)=\frac{1+(1-\theta)z}{1-\theta z}, |R(z)|\le 1, z\in\mathbb{R}^-$, 则

$$\begin{aligned}(\frac{1+(1-\theta)z}{1-\theta z})^2\le 1 &\Rightarrow (1+(1-\theta)z)^2\le(1-\theta z)^2\\ &\Rightarrow 1+2(1-\theta)z+(1-\theta)^2z^2\le 1-2\theta z+\theta^2z^2\\ &\Rightarrow 2z+z^2-2\theta z^2\le 0\\ &\Rightarrow (1-2\theta)z^2\le -2z\end{aligned} \tag{3}$$

1. 若 $\theta\in[\frac{1}{2},1], (1-2\theta)z^2\le 0\le -2z$, 此时方法 A-stable, 故 MOL 方法无条件稳定。
2. 若 $\theta\in[0,\frac{1}{2}], (1-2\theta)z^2\le -2z\Rightarrow z\ge\frac{-2}{1-2\theta}$, 因为 $\lambda\ge -\frac{4v}{h^2}$, 所以 $k\le\frac{h^2}{2v(1-2\theta)}$ 时 MOL 方法稳定。